

Keeping Type 2 Diabetes Patients Connected: Predicting 6 Month Follow-Up Gaps

Type 2 diabetes is a chronic condition that requires constant care. This means that patients need to regularly receive monitoring, lab work, and follow-up appointments. However, large health systems such as SVH need a way to know which patients are most likely to go too long without another hospital encounter. Therefore, the goal of our project is to identify Type 2 diabetes patients who are at higher risk of a delayed follow-up. We chose to measure follow-up gaps because they are both clinically meaningful and measurable in the data. Then, we narrowed down our scope to Type 2 diabetes because different conditions naturally have different follow-up schedules. What is a normal amount of time between encounters or check-ups in one condition might be more concerning for another condition, such as diabetes. By narrowing the scope, we are able to interpret our results in a significant and actionable way.

In making the model, we first define the beginning of the patient journey as the initial diabetes related encounter for the specific patient in the dataset. We then store encounters by date and calculate the maximum days between consecutive diabetes related encounters. This way, we are able to identify those who had a follow-up gap greater than 180 days. To ensure fairness, we also removed those who had under 365 days of total observation time and removed the first encounter year from the final model because, although statistically significant, it does not reflect real patient risk. After filtering, we found that 74.8% of Type 2 diabetes journeys had at least one gap longer than 180 days, with the median max gap being 258 days.

To produce meaningful insight, we built a random forest model to predict patients who were most likely to experience a 180-day follow-up gap. The model uses information such as first department specialty, first encounter type, visit type, age group, MyChart status, and several social determinants such as transportation, food insecurity, housing instability, financial strain, and stress to place patients into low, medium and high risk follow-up tiers with the risks being 61.9%, 79.9%, and 85% respectively.

Our proposed solution is to introduce general changes with specific protocols based on follow-up risk tiers. Firstly, volunteers should be hired to help in labs and offices to improve efficiency in wait times. Secondly, healthcare professionals should offer MyChart to patients. Thirdly, patients identified as having a social risk should be followed-up with after their encounter. Finally, patients deemed to be a part of the high-risk group should have their follow-up appointments booked before ending their encounter. This approach allows SVH to use follow up resources more effectively by focusing attention on patients who are most likely to experience long gaps.